

3.1 voda + led, vrženo na vroče železo

$$m_v = 0,8 \text{ kg}$$

$$m_l = 0,2 \text{ kg}$$

$$m_{Fe} = 1 \text{ kg}$$

$$T = ?$$

$$\Delta S = ?$$

topl. izoliran sistem

$$C_{p,v} = 4180 \frac{\text{J}}{\text{kg K}}$$

$$C_{p,Fe} = 450 \frac{\text{J}}{\text{kg K}}$$

$$q_t = 336 \frac{\text{kJ}}{\text{kg}}$$

$$T_{Fe} = 100^\circ\text{C}, T_0 = 0^\circ\text{C}$$

$$Q_{Fe} = m_{Fe} C_{p,Fe} (T_{Fe} - T_0) = 45000 \text{ J}$$

$$Q_l = m_l q_t = 67,2 \text{ kJ}$$

$$\rightarrow \underline{\underline{T = 0^\circ\text{C}}}$$

$$m'_l = \frac{Q_{Fe}}{q_t} = 0,134 \text{ kg (se stali)}$$

$$\Delta S \geq \int \frac{dQ}{T}$$

$$\Delta S = \Delta S_v + \Delta S_l + \Delta S_{Fe}$$

$$\Delta S_v = 0$$

$$\Delta S_l = \frac{m'_l q_t}{T_l} = 164,9 \frac{\text{J}}{\text{K}}$$

$$\Delta S_{Fe} = \int_{T_{Fe}}^{T_0} \frac{m_{Fe} C_{p,Fe} dT}{T} = m_{Fe} C_{p,Fe} \ln \frac{T_0}{T_{Fe}} = -140,4 \frac{\text{J}}{\text{K}}$$

$$\Delta S = 24,5 \frac{\text{J}}{\text{K}}$$

3.2) Podhl. voda (m) $T' = -20^\circ\text{C}$, vrženo na zmrzne ledu v topl. izd. pos.

$$\frac{m'}{m} \rightarrow \text{del vode, ki zmrzne}$$

$$\frac{m'}{m} = ?$$

$$\Delta S = \frac{\Delta S}{m} = ?$$

$$q_t = 336 \frac{\text{kJ}}{\text{kg}} \quad (0^\circ\text{C}, 10^5 \text{ Pa})$$

$$c_p^L = 2100 \frac{\text{J}}{\text{kg K}}$$

$$c_p^V = 4180 \frac{\text{J}}{\text{kg K}}$$

$$Q_v = m c_p^V (T - T')$$

$$Q_i = m' q_t$$

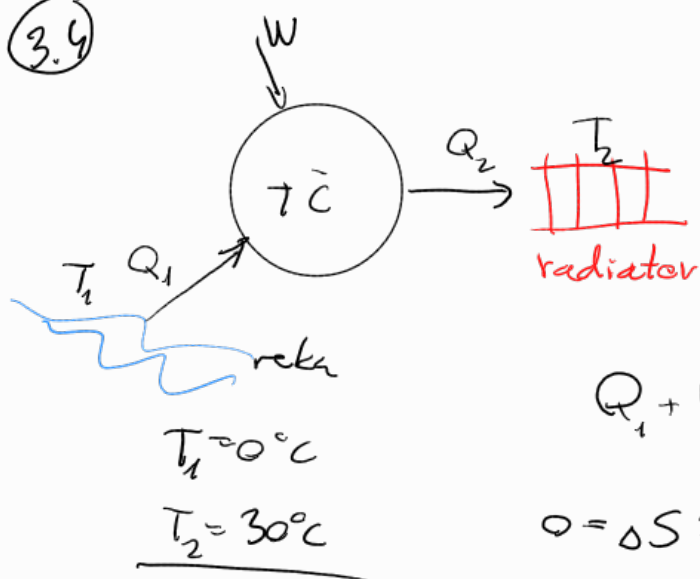
$$m c_p^V (T - T') = m' q_t$$

$$\frac{m'}{m} = \frac{20 \cdot 4180}{336000} = 0,25$$

$$\Delta S = \Delta S_v + \Delta S_i = m c_p^V \ln \frac{T}{T'} + m' q_t$$

$$\Delta S = \frac{\Delta S}{m} = c_p^V \ln \frac{T}{T'} - \frac{m'}{m} \frac{q_t}{T} = 11,8 \frac{\text{J}}{\text{kg K}}$$

3.4)



$$Q_1 + W = Q_2$$

$$0 = \Delta S \geq \int \frac{dQ}{T} = \frac{Q_1}{T_1} - \frac{Q_2}{T_2}$$

$$\frac{Q_2 - W}{T_1} = \frac{Q_1}{T_1} \leq \frac{Q_2}{T_2} \rightarrow Q_2 \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \leq \frac{W}{T_1}$$

$$\frac{W}{Q_2} \geq \frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2}$$

$\sim 0,1$

3.6 Carnotov topl. str.

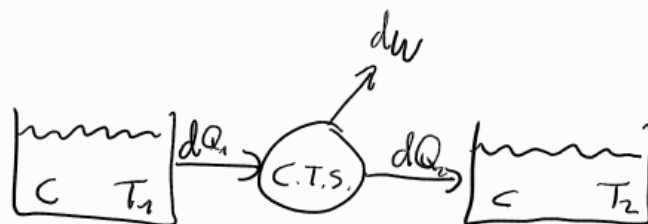
$$T_1' = 200^\circ\text{C}$$

$$T_2' = 150^\circ\text{C}$$

$$C = 5 \frac{\text{MJ}}{\text{K}}$$

$$T_k = ?$$

$$W = ?$$



$$0 = dS = \frac{dQ_1}{T_1} - \frac{dQ_2}{T_2} = 0$$

$$-\frac{\cancel{C}dT_1}{T_1} = \frac{\cancel{C}dT_2}{T_2}$$

$$\int_{T_k}^{T_1'} \frac{dT_1}{T_1} = \int_{T_2'}^{T_k} \frac{dT_2}{T_2}$$

$$\ln \frac{T_1'}{T_k} = \ln \frac{T_k}{T_2'}$$

$$T_k = \sqrt{T_1' T_2'} = 447,3 \text{ K} = 174,3^\circ\text{C}$$

$$dW = dQ_1 - dQ_2 = -CdT_1 - CdT_2$$

$$W = -C(T_k - T_1) - C(T_k - T_2) =$$

$$= C(T_1 - 2T_k + T_2) =$$

$$= 7 \text{ MJ}$$